1.05 Linear Modelling

Worked Solutions

1 a)

$y = \frac{1}{2}x$

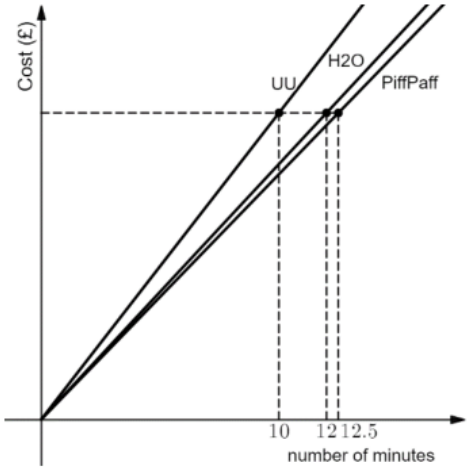
1 b)

$y = 1.2x$

1 c)

$y = 1.29x$

2 a)



2 b)

UU: $C = 0.3t$

H_2O: $C = 0.25t$

Piffpaff: $C = 0.24t$

3 a)

$m = \frac{9-8}{1} \Rightarrow y = x + 8$

3 b)

$m = \frac{4-2}{5} \Rightarrow y = \frac{2}{5}x + 2$

3 c)

$m = \frac{7-5}{3} \Rightarrow y = \frac{2}{3}x + 5$

4 a)

$m = \frac{19-10}{5-2} = 3$

$10 = 3 \times 2 + c \Rightarrow c = 4$

$y = 3x + 4$

4 b)

$m = \frac{5.5-2.5}{7-1} = 0.5$

$2.5 = 0.5 \times 1 + c \Rightarrow c = 2$

$y = 0.5x + 2$

4 c)

$m = \frac{12-7}{15-3} = \frac{5}{12}$

$7 = \frac{5}{12} \times 3 + c \Rightarrow c = \frac{23}{4}$

$y = \frac{5}{12}x + \frac{23}{4}$

5 a)

Answers similar to $d = 2.8t + 0.2$

5 b)

Answers close to 7.2 feet

5 c)

Depth per minute (feet per minute)

5 d)

Vertical intercept is 0.2 which is the theoretical time that the penguins will dive in 0 minutes

5 e)

This is not possible so the model is not valid for low values.

6 a)

$x = 68$

$T = -0.0082 \times 68 + 10.2 = 9.6424$

$\therefore$ about 9.6 seconds

6 b)

The 10.2 represents a theoretical time for how long the gold medallist ran for in 1960.

The -0.0082 represents the rate of change in time per year, that is, for every 1 year that passes, the time to complete the 100m race goes down by 0.0082 seconds.

6 c)

As the years pass, the times are getting smaller by 0.0082 a year, so eventually it will become too quick for it to be possible for a human to achieve it. Eventually (in 3204) the model will predict a negative race time!

7 a)

Zelacoach: £800 cost for hiring the coach

Stagebus: £500 cost for hiring the coach

7 b)

Zelacoach: $1700 = 220 \times m + 800$

$\Rightarrow C = \frac{45}{11}m + 800$

Stagebus: $2000 = 220 \times m + 500$

$\Rightarrow C = \frac{75}{11}m + 500$

7 c)

$\frac{45}{11}m + 800 = \frac{75}{11}m + 500 \Rightarrow 300 = \frac{30}{11}m$

$\therefore m = 110$

P is $(110, 1250)$

When the number of miles is 110, both companies charge the same amount, £1250

8)

Supply:

$m = \frac{3.52-1.45}{168-30} = \frac{3}{200}$

$\Rightarrow P - 1.45 = \frac{3}{200}(x - 30) \Rightarrow P = \frac{3}{200}x + 1$

Demand:

$m = \frac{4.1-1.5}{36-140} = -\frac{1}{40}$

$\Rightarrow P - 4.1 = -\frac{1}{40}(x - 36) \Rightarrow P = -\frac{1}{40}x + 5$

$\frac{3}{200}x + 1 = -\frac{1}{40}x + 5 \Rightarrow \frac{1}{25}x = 4 \therefore x = 100$

$P = 2.5$

$(100, 2.5)$

When ice creams are sold at £2.50, 100 ice creams are sold. For a higher price point, the demand goes down as does the number sold. For a lower price point, the demand goes up as does the number sold.

9 a)

At $(4, 170)$

$y = 30x + k$ gives $170 = 30 \times 4 + k$ so $k = 50$

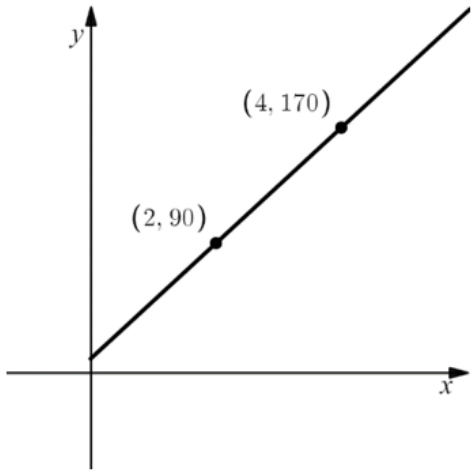
£50 call-out charge

9 b)

$\frac{170-90}{4-2} = 40$

9 c)

Ohms Electrical charges £40 for every hour worked starting with a minimum charge of £90 but with 2 hours work included:



$y - 90 = 40(x - 2) \Rightarrow y - 90 = 40x - 80$

$\Rightarrow y = 40x + 10$

Charge at 0 hours would be £10

10 a)

Profit/loss = income − costs

When 500 muffins are made and sold:

$300 = 1000 - 500C$

$\therefore C = 700$

When 100 muffins are made and sold:

$-50 = 200 - 100C$

$\therefore C = 250$

Let F be the fixed cost to the bakery and m be the variable cost to the bakery.

$500m + F = 700$

$100m + F = 250$

Solving gives:

$m = 1.125$ and $F = 137.5$

$\therefore C = 1.125x + 137.5$

10 b)

Consider when the sales are the same as the cost:

$2x = 1.125x + 137.5 \Rightarrow n = \frac{1100}{7} = 157.14\ldots$

So need to make at least 158 muffins to make a profit.